



Final Project ECSE 500

Mathematical Foundations of Systems (McGill University)



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1. Let $a < b$ in $\mathbb{R}$. Show that the space $W[a, b]$ of continuously differentiable functions on $[a, b]$ with values in $\mathbb{R}$, is an inner product space with respect to pointwise addition $((f+g)(t) = f(t)+g(t))$ and scalar multiplication $((\alpha f)(t) = \alpha f(t))$, and inner product

$$\langle f, g \rangle = \int_a^b f(t)g(t) + f'(t)g'(t)dt$$

Solution:

We can easily prove that $W[a, b]$ is a vector space with the two operations. (It is obvious that point-wise addition of two continuous differentiable functions and scalar multiplication of a continuous differentiable are both continuous differentiable functions.) The remaining things we need to prove is that this kind of inner product satisfies the three conditions.

(1) $\forall f(t) \in W[a, b]$, we have $f(t)^2 \geq 0$ and $f'(t)^2 \geq 0$, $\forall t \in [a, b]$, so

$$\langle f, f \rangle = \int_a^b f(t)^2 + f'(t)^2 dt \geq 0$$

the equality holds if and only if $f(t) = 0, t \in [a, b]$.

(2) Linearity in the first slot.

$\forall f(t), g(t), h(t) \in W[a, b]$, and $\alpha \in \mathbb{R}$, we have

$$\begin{aligned} \langle f + g, h \rangle &= \int_a^b (f(t) + g(t))h(t) + (f'(t) + g'(t))h'(t)dt \\ &= \int_a^b f(t)h(t) + f'(t)h'(t)dt + \int_a^b g(t)h(t) + g'(t)h'(t)dt \\ &= \langle f, h \rangle + \langle g, h \rangle \end{aligned}$$

and

$$\begin{aligned} \langle \alpha f, h \rangle &= \int_a^b (\alpha f(t))h(t) + (\alpha f'(t))h'(t)dt \\ &= \alpha \int_a^b f(t)h(t) + f'(t)h'(t)dt \\ &= \alpha \langle f, h \rangle \end{aligned}$$

which gives the linearity in the first slot.

(3) Conjugate symmetry turns into symmetry since it is a real value function space.

$\forall f(t), g(t) \in W[a, b]$, we have

$$\begin{aligned} \langle f, g \rangle &= \int_a^b f(t)g(t) + f'(t)g'(t)dt \\ &= \int_a^b g(t)f(t) + g'(t)f'(t)dt \\ &= \langle g, f \rangle \end{aligned}$$

which gives symmetry.

Combine(1), (2), and (3), we derive that it is an inner product space.

2. Prove that in $W[0, 1]$

$$\langle f, \cosh \rangle = f(1) \sinh 1$$

and deduce that

$$\{f \in W[0, 1] \mid f(1) = 0\}$$

is a closed subspace of $W[0, 1]$.

Solution:

Assume we have the definition of inner product in the first question. For $f(t) \in W[0, 1]$, we have

$$\begin{aligned} \langle f, \cosh \rangle &= \int_0^1 f(t) \cosh(t) + f'(t) \cosh'(t) dt \\ &= \int_0^1 f(t) \cosh(t) + f'(t) \sinh(t) dt \\ &= \frac{1}{2} \int_0^1 (f(t)(e^t + e^{-t}) + f'(t)(e^t - e^{-t})) dt \\ &= \frac{1}{2} \left(\int_0^1 e^t (f(t) + f'(t)) dt + \int_0^1 e^{-t} (f(t) - f'(t)) dt \right) \\ &= \frac{1}{2} ((e^t f(t))|_0^1 + (-e^{-t} f(t))|_0^1) \\ &= f(1) \cdot \frac{e^1 - e^{-1}}{2} \\ &= f(1) \sinh 1 \end{aligned}$$

which gives the equation.

Since any inner product space could induce a norm, we can consider any convergent sequence $\{f_n\}$ in $\mathcal{S}$, and suppose the limit of the sequence is f , which is not necessarily belongs to $\mathcal{S}$. We need to show that it belongs to $\mathcal{S}$.

Use Cauchy-Schwartz to upper bound difference of inner product, we have

$$|\langle f_n, \cosh \rangle - \langle f, \cosh \rangle| = |\langle f_n - f, \cosh \rangle| \leq \|f_n - f\| \|\cosh\|$$

notice that $\langle f_n, \cosh \rangle = 0$ since $f_n \in \mathcal{S}$, so the inequality turns into

$$|\langle f, \cosh \rangle| \leq \|f_n - f\| \|\cosh\|$$

then take limit $n \rightarrow \infty$ on both side, right-hand side equals to zero, so left-hand side also equals to zero. And, $\langle f, \cosh \rangle = f(1) \sinh(1) = 0$, so $f(1) = 0$, which means that $f \in \mathcal{S}$. So every limit point of $\mathcal{S}$ is an element of $\mathcal{S}$, so $\mathcal{S}$ is a closed set.

3. c_0 denotes the subspace of l^∞ comprising all sequences $\{x_n\}$ which converge to zero as $n \rightarrow \infty$. Prove that c_0 is closed in l^∞ (with respect to $\|\cdot\|_\infty$).

Solution:

Consider any convergent sequence of sequences $\{x_n^1\}, \{x_n^2\}, \dots, \{x_n^k\}, \dots$. Every sequence $\{x_n^j\}$ converges to 0, which means that $\forall j \in \{1, 2, \dots\}$

$$\forall \varepsilon \geq 0, \exists N \in \mathbb{N}, s.t. \forall n > N, |x_n^j| < \varepsilon$$

and assume the sequence of sequences converges into a sequence $\{x_n\}$.

List all of the entries,

$$\begin{array}{c} x_1^1, x_2^1, \dots, x_n^1 \dots \rightarrow 0 \\ x_1^2, x_2^2, \dots, x_n^2 \dots \rightarrow 0 \\ \dots \dots \dots \dots \\ x_1^k, x_2^k, \dots, x_n^k \dots \rightarrow 0 \\ \vdots \quad \vdots \quad \quad \vdots \\ \Downarrow \\ x_1, x_2, \dots, x_n, \dots \end{array}$$

Because $\{x_n\}$ is a limit of the sequences with respect to $\|\cdot\|_\infty$, so $\forall \varepsilon/2 > 0, \exists K \in \mathbb{N}, s.t. \forall k \geq K$, we have

$$\|x_n^k - x_n\|_\infty < \varepsilon/2$$

which is

$$\sup_{n \in \{1, 2, \dots\}} |x_n^k - x_n| < \varepsilon/2 \quad (3.1)$$

And for any fixed k , the sequence $\{x_n^k\}$ converges to 0. So for the $\varepsilon/2 > 0, \exists N \in \mathbb{N}, s.t. \forall n > N$, we have

$$|x_n^k| < \varepsilon/2 \quad (3.2)$$

Since the supremum of a set would not increase after deleting some elements, we have

$$\sup_{n \in \{N, N+1, \dots\}} |x_n - x_n^k| = \sup_{n \in \{N, N+1, \dots\}} |x_n^k - x_n| \leq \sup_{n \in \{1, 2, \dots\}} |x_n^k - x_n| < \varepsilon/2 \quad (3.3)$$

Due to (3.2), we have

$$\sup_{n \in \{N, N+1, \dots\}} |x_n^k| \leq \varepsilon/2 \quad (3.4)$$

Add (3.3) and (3.4), get

$$\begin{aligned} \sup_{n \in \{N, N+1, \dots\}} |x_n| &\leq \sup_{n \in \{N, N+1, \dots\}} |x_n - x_n^k| + |x_n^k| \\ &\leq \sup_{n \in \{N, N+1, \dots\}} |x_n - x_n^k| + \sup_{n \in \{N, N+1, \dots\}} |x_n^k| \\ &< \varepsilon \end{aligned}$$

which means that ε is an upper bound of $|x_n|, n \in \{N, N+1, \dots\}$. By definition, $\{x_n\}$ converges to 0, so $\{x_n\}$ is also an element in c_0 . So c_0 is closed in l^∞ .

4. Prove that l^∞ with the supremum norm $\|\cdot\|_\infty$ is complete.

Solution: Consider any Cauchy sequence in l^∞ : $\{x_n^1\}, \{x_n^2\}, \dots, \{x_n^k\}, \dots$. Because it is a Cauchy sequence, so $\forall \varepsilon > 0, \exists K \in \mathbb{N}, s.t. \forall p, l > K$, we have

$$\sup_{i \in \{1, 2, \dots\}} |x_i^p - x_i^l| < \varepsilon$$

so, for any fixed i , we have

$$|x_i^p - x_i^l| < \varepsilon \quad (4.1)$$

which means for any fixed i , $\{x_i^j\}, j \in \{1, 2, \dots\}$ is a Cauchy sequence. Since the complex plane is a complete metric space, I assume the sequence converges to $x_i, i = 1, 2, \dots$. So, we have already derived a candidate limit $\{x_n\}$, but it is not necessarily an element of l^∞ , we need to prove that.

Take the limit $l \rightarrow \infty$ on both sides of (4.1), we have

$$|x_i^p - x_i| \leq \varepsilon, p > K \quad (4.2)$$

Since $\forall p \in \{1, 2, \dots\}, \{x_n^p\} \in l^\infty$, so $\sup_{i \in \{1, 2, \dots\}} |x_i^p| \leq c_p$, where c_p is a constant. So we have

$$|x_i^p| \leq c_p, i \in \{1, 2, \dots\} \quad (4.3)$$

Add (4.2) and (4.3), we get

$$|x_i| \leq |x_i - x_i^p| + |x_i^p| \leq |x_i^p - x_i| + |x_i^p| \leq \varepsilon + c_p$$

the inequality holds for every $i \in \{1, 2, \dots\}$, so $\{x_n\}$ is also an element in $l \rightarrow \infty$.

From (4.2), we can get

$$\sup_{i \in \{1, 2, \dots\}} |x_i^p - x_i| \leq \varepsilon$$

By definition, $\{x_n\}$ is a limit of the sequence $\{x_n^1\}, \{x_n^2\}, \dots, \{x_n^k\}, \dots$ with respect to $\|\cdot\|_\infty$. So we have derived that any Cauchy sequence in l^∞ converges to an element in l^∞ , so l^∞ is complete.

5. A *Volterra operator* V on $L^2(0, 1)$ is an operator of integration such as the one shown below

$$(Vx)(t) = \int_0^t x(s)ds \quad 0 \leq t \leq 1$$

- (a) Show that V is a bounded linear operator and that $\|V\| \leq \frac{1}{\sqrt{2}}$.
- (c) An operator is said to have *rank 1* if its range is one-dimensional. Show that $V + V^*$ has rank 1.

Hint:

In part (a) use the Schwartz inequality.

Solution:

(a) Use Cauchy-Schwartz inequality to separate the L^2 norm of $x(t)$ from the result of the operation:

$$\begin{aligned} |(Vx)(t)| &= \left| \int_0^t x(s)ds \right| \leq \int_0^t 1 \cdot |x(s)|ds \\ &\leq \left(\int_0^t 1ds \right)^{\frac{1}{2}} \left(\int_0^t |x(s)|^2 ds \right)^{\frac{1}{2}} \\ &\leq \sqrt{t} \|x(t)\|_{L^2(0,1)} \end{aligned}$$

then, calculate the L^2 norm of the result of the operation:

$$\|(Vx)(t)\|_{L^2(0,1)} = \left(\int_0^1 |(Vx)(t)|^2 dt \right)^{\frac{1}{2}} \leq \left(\int_0^1 t dt \right)^{\frac{1}{2}} \|x(t)\|_{L^2(0,1)} = \frac{1}{\sqrt{2}} \|x(t)\|_{L^2(0,1)}$$

so, we have already showed that V is bounded because there exist $c = \frac{1}{\sqrt{2}}$, such that $\forall x(t) \in L^2(0, 1)$, the following inequality is satisfied:

$$\|Vx\| \leq c\|x\|$$

and the norm of V is

$$\|V\| = \sup_{\substack{x \in L^2(0,1) \\ x \neq 0}} \frac{\|Tx\|}{\|x\|} \leq \sup_{\substack{x \in L^2(0,1) \\ x \neq 0}} c = c = \frac{1}{\sqrt{2}}$$

- (c) An adjoint operator should satisfy $\forall x, y \in L^2(0, 1)$,

$$\langle Vx, y \rangle = \langle x, V^*y \rangle$$

Interchange order of integration to separate $y(t)$ in the L.H.S.

$$\langle Vx, y \rangle = \int_0^1 \left(\int_0^t x(s)ds \right) y(t)dt = \int_0^1 \int_s^1 y(t)dt x(s)ds$$

By comparing the above result and the result of $\langle x, V^*y \rangle$, we derive that $(V^*y)(t) = \int_t^1 y(s)ds$. So,

$$(V + V^*)(x)(t) = \int_0^1 x(s)ds$$

which is a constant, so any two elements in the range space is linearly dependent, and the range not only contain the zero vector. Thus, we can say that the range space is of rank 1.

6. Let T be a bounded linear operator of rank 1 on a Hilbert space H , and let ψ be a non-zero vector in the range of T .

- (a) Show that there exists $\varphi \in H$ such that

$$\forall x \in H \quad Tx = \langle x, \varphi \rangle \psi$$

and that

$$\|T\| = \|\varphi\| \|\psi\|$$

- (b) Show that

$$T^2 = \langle \psi, \varphi \rangle T$$

deduce that

$$\langle \psi, \varphi \rangle \neq 1 \quad \Rightarrow \quad I - T \text{ is a bijection on } H$$

and find $(I - T)^{-1}$

- (c) What is T^* ?

Solution:

(a) Because the range space of the linear operator is of rank 1 and ψ is a non-zero vector in it, so $\forall x \in H, \exists c_x \in \mathbb{R}$, such that $Tx = c_x \psi$. Regard $f : x \mapsto c_x$ as a functional, because T is linear and bounded, then the functional f is also linear and bounded.

$$T(ax + by) = aT(x) + bT(y) = (ac_x + bc_y)\psi, \quad c_{ax+by} = ac_x + bc_y$$

And, using the Riesz's Theorem, we can say that there $\exists \varphi \in H$, such that $\forall x \in H, f(x) = c_x = \langle x, \varphi \rangle$, which is equivalent to $Tx = \langle x, \varphi \rangle \psi$. Also, the Riesz's Theorem gives that $\|f\| = \|\varphi\|$.

Consider the norm of the linear operator,

$$\begin{aligned} \|T\| &= \sup_{\substack{x \in H \\ x \neq 0}} \frac{\|Tx\|}{\|x\|} = \sup_{\substack{x \in H \\ x \neq 0}} \frac{\|\langle x, \varphi \rangle \psi\|}{\|x\|} = \sup_{\substack{x \in H \\ x \neq 0}} \frac{|\langle x, \varphi \rangle| \|\psi\|}{\|x\|} \\ &= \|\psi\| \sup_{\substack{x \in H \\ x \neq 0}} \frac{|\langle x, \varphi \rangle|}{\|x\|} = \|\psi\| \|f\| = \|\psi\| \|\varphi\| = \|\varphi\| \|\psi\| \end{aligned}$$

- (b) From the targeting equation we know that ψ also lives in the same Hilbert space H , so we should know that the range of T also belongs to the Hilbert space H .

$\forall x \in H$, we have

$$T^2x = T\langle x, \varphi \rangle \psi = \langle x, \varphi \rangle T\psi = \langle x, \varphi \rangle \langle \psi, \varphi \rangle \psi = \langle \psi, \varphi \rangle \langle x, \varphi \rangle \psi$$

and,

$$\langle \psi, \varphi \rangle Tx = \langle \psi, \varphi \rangle \langle x, \varphi \rangle \psi$$

so $T^2 = \langle \psi, \varphi \rangle T$.

i) If $\varphi = 0$, then $T = 0$, $I - T = I$ is a bijection on H , and $\forall x \in H$

$$\begin{aligned}(I + T)(I - T)x &= (I + T)(x - Tx) = (I + T)x = Ix = x \\ (I - T)(I + T)x &= (I - T)(x + Tx) = (I - T)x = Ix = x\end{aligned}$$

so $(I - T)^{-1} = I + T$.

ii) If $\varphi \neq 0$, then if $(I - T)x_1 = (I - T)x_2$ for some $x_1, x_2 \in H$, then

$$\begin{aligned}(I - T)x_1 &= (I - T)x_2 \Leftrightarrow x_1 - x_2 = T(x_1 - x_2) \Leftrightarrow x_1 - x_2 = \langle x_1 - x_2, \varphi \rangle \psi \\ &\Rightarrow \langle x_1 - x_2, \varphi \rangle = \langle x_1 - x_2, \varphi \rangle \langle \psi, \varphi \rangle \\ &\Rightarrow \langle x_1 - x_2, \varphi \rangle = 0 \Rightarrow x_1 - x_2 = 0 \Rightarrow x_1 = x_2\end{aligned}$$

So $I - T$ is an injection. And $\forall y$ in the range of T , find a $x \in H$, such that $(I - T)x = y$, indeed, take

$$x = y + \frac{1}{1 - \langle \psi, \varphi \rangle} \langle y, \varphi \rangle \psi$$

then

$$\begin{aligned}(I - T)x &= (I - T) \left(y + \frac{1}{1 - \langle \psi, \varphi \rangle} \langle y, \varphi \rangle \psi \right) \\ &= \left(y + \frac{1}{1 - \langle \psi, \varphi \rangle} \langle y, \varphi \rangle \psi \right) - T \left(y + \frac{1}{1 - \langle \psi, \varphi \rangle} \langle y, \varphi \rangle \psi \right) \\ &= y + \frac{1}{1 - \langle \psi, \varphi \rangle} \langle y, \varphi \rangle \psi - \langle y, \varphi \rangle \psi - \frac{1}{1 - \langle \psi, \varphi \rangle} \langle y, \varphi \rangle \langle \psi, \varphi \rangle \psi \\ &= y + \langle y, \varphi \rangle \psi - \langle y, \varphi \rangle \psi = y\end{aligned}$$

which means that $I - T$ is a sujection on H . Then $I - T$ is both an injection and sujection, so it is a bijection on H . And $(I - T)^{-1} = I + \frac{1}{1 - \langle \psi, \varphi \rangle} T$, indeed, $\forall x \in H$

$$\begin{aligned}(I - T)(I + \frac{1}{1 - \langle \psi, \varphi \rangle} T)x &= (I - T)(x + \frac{1}{1 - \langle \psi, \varphi \rangle} \langle x, \varphi \rangle \psi) \\ &= x + \frac{1}{1 - \langle \psi, \varphi \rangle} \langle x, \varphi \rangle \psi \\ &\quad - \langle x, \varphi \rangle \psi - \frac{1}{1 - \langle \psi, \varphi \rangle} \langle x, \varphi \rangle \langle \psi, \varphi \rangle \psi \\ &= x \\ (I + \frac{1}{1 - \langle \psi, \varphi \rangle} T)(I - T)x &= (I + \frac{1}{1 - \langle \psi, \varphi \rangle} T)(x - \langle x, \varphi \rangle \psi) \\ &= x - \langle x, \varphi \rangle \psi \\ &\quad + \frac{1}{1 - \langle \psi, \varphi \rangle} \langle x, \varphi \rangle \psi - \frac{1}{1 - \langle \psi, \varphi \rangle} \langle x, \varphi \rangle \langle \psi, \varphi \rangle \psi \\ &= x\end{aligned}$$

So, $(I - T)^{-1} = I + \frac{1}{1 - \langle \psi, \varphi \rangle} T$.

(c) $T^*y = \langle y, \psi \rangle \varphi$, $\forall y$ in the range of T .

Because the range of T also belongs to the same Hilbert space H , so $\forall x, y \in H$, we have

$$\langle Tx, y \rangle = \langle \langle x, \varphi \rangle \psi, y \rangle = \langle x, \varphi \rangle \langle \psi, y \rangle,$$

$$\langle x, T^*y \rangle = \langle x, \langle y, \psi \rangle \varphi \rangle = \overline{\langle y, \psi \rangle} \langle x, \varphi \rangle = \langle \psi, y \rangle \langle x, \varphi \rangle = \langle x, \varphi \rangle \langle \psi, y \rangle$$

so, $T^*y = \langle y, \psi \rangle \varphi$, $\forall y$ in the range of T .

7. Let $\mathcal{D}$ be the space of differentiable functions $f \in L^2(-\infty, \infty)$ such that $f' \in L^2(-\infty, \infty)$. Then

$$\frac{d}{dt} : \mathcal{D} \rightarrow L^2(-\infty, \infty)$$

is a linear operator. Show that it is unbounded with respect to the L^2 norm in $\mathcal{D}$

$$\sqrt{\int_{-\infty}^{\infty} |f(t)|^2 dt}$$

Now, suppose that $\mathcal{D}$ is equipped with the norm induced by the following scalar product:

$$\langle f, g \rangle = \int_{-\infty}^{\infty} f(t)g(t) + f'(t)g'(t)dt$$

Show that with this new norm in $\mathcal{D}$, the differential operator defined above is in fact bounded.

Solution:

1) Now consider a series of functions on $L^2(-\infty, \infty)$ that their derivatives' L^2 norms have unbounded limit.

Let $f_n(x) = \sqrt{n}e^{-n^2x^2}$, they have finite L^2 norm, indeed,

$$\begin{aligned} \int_{-\infty}^{+\infty} |f_n(x)|^2 dx &= \int_{-\infty}^{+\infty} ne^{-2n^2x^2} dx = n \sqrt{\int_0^{2\pi} \int_0^{+\infty} re^{-2n^2r^2} dr d\theta} \\ &= n \sqrt{\int_0^{2\pi} -\frac{1}{4n^2} e^{-2n^2r^2} \Big|_0^{+\infty} d\theta} \\ &= n \sqrt{\frac{2\pi}{4n^2}} = \sqrt{\frac{\pi}{2}} \end{aligned}$$

However, limit of L^2 norms of their derivatives are unbounded,

$$\begin{aligned} \int_{-\infty}^{+\infty} |f'_n(x)|^2 dx &= \int_{-\infty}^{+\infty} 4n^5 x^2 e^{-2n^2x^2} dx = \int_{-\infty}^{+\infty} 4n^5 x d\left(-\frac{1}{4n^2} e^{-2n^2x^2}\right) \\ &= -4n^5 \int_{-\infty}^{+\infty} -\frac{1}{4n^2} e^{-2n^2x^2} dx = n^2 \int_{-\infty}^{+\infty} ne^{-2n^2x^2} dx = n^2 \sqrt{\frac{\pi}{2}} \end{aligned}$$

So for every $M > 0$, there exist an $N = M$, such that $\forall n > N$

$$\frac{\|f'_n(x)\|_{L^2(-\infty, \infty)}}{\|f_n(x)\|_{L^2(-\infty, \infty)}} = n > N = M$$

By definition, the differentiation operator with respect to L^2 norm is unbounded.

2) $\forall f \in \mathcal{D}$, $\|f\|_{L^2(-\infty, \infty)}$ is bounded and $\|f'\|_{L^2(-\infty, \infty)}$ is also bounded. So f' also belongs to $\mathcal{D}$, then $\|f''\|_{L^2(-\infty, \infty)}$ is also bounded. Thus we assume that

$\forall f \in \mathcal{D}, \|f\|_{L^2(-\infty, \infty)} \leq N, \|f'\|_{L^2(-\infty, \infty)} \leq N_1, \|f''\|_{L^2(-\infty, \infty)} \leq N_2.$

Consider the norm of the operator,

$$\begin{aligned} \left\| \frac{d}{dt} \right\| &= \sup_{\substack{\forall f \in \mathcal{D} \\ f \neq 0}} \frac{\left\| \frac{df}{dt} \right\|}{\|f\|} = \sup_{\substack{\forall f \in \mathcal{D} \\ f \neq 0}} \frac{\sqrt{\int_{-\infty}^{\infty} f'(t)^2 + f''(t)^2 dt}}{\sqrt{\int_{-\infty}^{\infty} f(t)^2 + f'(t)^2 dt}} \\ &= \sup_{\substack{\forall f \in \mathcal{D} \\ f \neq 0}} \frac{\sqrt{\|f'\|_{L^2(-\infty, \infty)}^2 + \|f''\|_{L^2(-\infty, \infty)}^2}}{\sqrt{\|f\|_{L^2(-\infty, \infty)}^2 + \|f'\|_{L^2(-\infty, \infty)}^2}} \end{aligned}$$

where the above result should be bounded since every term is bounded and the denominator is not equal to zero.